Introduction to Artificial Intelligence and Machine Learning (CSN11601)

LAB TASK

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Q1. **Scenario: Sorting Shipment Boxes**

You are working as a software trainee for a logistics company. The company receives a daily list of shipment box weights that need to be sorted for loading into delivery vans. The sorting should help in organizing lighter boxes first to optimize loading and space.
Depending on the van type, the sorting method changes:

- For Type A vans, the boxes need to be sorted using the Bubble Sort algorithm.
- For Type B vans, the boxes need to be sorted using the Selection Sort algorithm.

You are provided a list of box weights (integers), the number of boxes, and the type of van.
Your task is to:
Take input from the user:

- Number of boxes (N)
- Box weights (N space-separated integers)
- Van type (A or B)

Sort the list using:

- Bubble Sort if van type is 'A'
- Selection Sort if van type is 'B'

Display the sorted list.
Constraints:
Do NOT use any functions (i.e., no def).
Use only loops, conditionals, and lists.
Sample Input:
Enter number of boxes: 5
Enter box weights: 45 12 78 34 23
Enter van type (A or B): A
Sample Output:
Sorted box weights: 12 23 34 45 78

A1. CODE:

```
#Name : Aayansh Niranjan
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n=int(input("Enter number of boxes: "))
str_boxes=list(input("Enter box weights: ").split())
boxes=[]
for i in str_boxes:
        boxes.append(int(i))
van_type=input("Enter van type (A or B): ")

if van_type=="A":
        for i in range(n-1):
                flag=True
                for j in range(n-i-1):
                        if boxes[j]>boxes[j+1]:
                                boxes[j],boxes[j+1]=boxes[j+1],boxes[j]
                                flag=False
                if flag:
                        break

elif van_type=="B":
        for i in range(n-1):
                min_idx=i
                for j in range(i+1,n):
                        if boxes[j]<boxes[min_idx]:
                                min_idx=j
                boxes[i],boxes[min_idx]=boxes[min_idx],boxes[i]

print("Sorted box weights:",end=" ")
for w in boxes:
   print(w,end=" ")
print()
```

OUTPUT:

```
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Lab_Task2$ python3 p1.py
Enter number of boxes: 4
Enter box weights: 12 120 36 6
Enter van type (A or B): A
Sorted box weights: 6 12 36 120
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Lab_Task2$ python3 p1.py
Enter number of boxes: 5
Enter box weights: 9 3 88 4 69
Enter van type (A or B): B
Sorted box weights: 3 4 9 69 88
```

Q2. **Scenario: Hospital Patient Queue**

In a hospital emergency ward, each patient is assigned a priority number when they arrive.

- The lower the priority number, the higher the urgency of treatment.
- As patients arrive one by one, the hospital staff needs to insert them into the queue at the correct position so that patients are always in order of priority.

You are asked to write a program using Insertion Sort to maintain the queue in ascending order of priority numbers.

Requirements

1. Input should contain the number of patients, followed by their Patient ID and Priority number.
2. Insert each patient into the queue using Insertion Sort logic.
3. Finally, display the sorted patient queue from highest priority to lowest priority.

Input Format

n
PatientID1 Priority1
PatientID2 Priority2
PatientIDn Priorityn

Output Format

Queue after sorting by priority:
PatientID Priority
PatientID Priority

Example

Input
5
P101 4
P102 2
P103 5
P104 1
P105 3
Output
Queue after sorting by priority:
P104 1
P102 2
P105 3
P101 4
P103 5

Implement the program using insertion sort only (no built-in sort functions).

A2. CODE:

```
#Name: Aayansh Niranjan
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n=int(input())
patients=[]
for i in range(n):
    data = input().split()
    patient_id = data[0]
    priority = int(data[1])
    patients.append([patient_id,priority])
    j=i
    while j > 0 and patients[j][1] < patients[j-1][1]:
            patients[j-1],patients[j]=patients[j],patients[j-1]
            j-=1
print("Queue after sorting by priority:")
```

OUTPUT:

```
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Lab_Task2$ python p2.py
5
P101 4
P102 5
P103 2
P104 1
P105 6
Queue after sorting by priority:
P104 1
P103 2
P101 4
P102 5
P105 6
```